Predicting Hotspot Emergence in Riau Province. A Multimodal Spatiotemporal Model Design for Seven-Days Early Warning

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Abstract

Fires in the tropical peatland ecosystems of Riau produce disproportionate carbon emissions and toxic haze because they tend to become subterranean smoldering fires. Existing frameworks are largely reactive and focus on post-event detection. We present a pre-implementation research design for identifying cells at elevated risk of hotspot emergence within the following seven days, on a 5 km $\times$ 5 km grid across Riau. The task is formulated as binary classification per cell-day rather than count regression or prediction of burned area and spread direction. Six data sources are fused, namely FIRMS VIIRS 375 m as the label source, ERA5-Land, CHIRPS v3 in its *sat* variant, Sentinel-1 SAR, Google Dynamic World, and a peatland map. Two feature regimes are compared. The *environmental* regime excludes fire history and carries the primary scientific contribution, whereas the *operational* regime includes fire history and establishes a practical performance ceiling. This separation is designed to distinguish the predictive power of environmental conditions from hotspot persistence effects, which frequently dominate and obscure model interpretation in prior work. The primary evaluation uses a temporal split with training on 2019 to 2022 and testing on 2023, supplemented by a spatial block robustness check with a seven-cell buffer. The contribution of this document is an auditable design framework comprising a precise labeling rule, a ten-item leakage checklist, and explicit claim boundaries that prevent the stated findings from exceeding what the method can demonstrate. This work evaluates per-cell risk classification rather than full-map forecasting.

1 Introduction

The province of Riau, Indonesia, represents a critical geographic domain for wildfire vulnerability, primarily due to its extensive tropical peatland ecosystems [1]. Peatland fires in this region are particularly hazardous because they frequently transition into subterranean smoldering fires, which release disproportionately massive amounts of carbon emissions and persistent toxic haze compared to standard surface fires [2]. Historically, severe dry spells often exacerbated by climatic anomalies such as the El Niño-Southern Oscillation (ENSO) drastically lower the water table in these peatlands, creating highly combustible environmental conditions [3]. Consequently, establishing a proactive early warning system that accurately predicts the emergence of hotspots prior to ignition is operationally imperative to facilitate effective, localized mitigation efforts [4].

Despite the urgent need for pre-event forecasting, contemporary wildfire management frameworks predominantly rely on reactive, post-event detection methodologies. High-resolution satellite instruments, such as the FIRMS VIIRS 375 m sensor, excel at identifying active thermal anomalies but offer limited utility for anticipatory resource deployment [5]. To address this operational gap, this research formulates a strictly binary spatiotemporal classification task targeting a 7-day forecasting window across a granular 5 km $\times$ 5 km spatial grid. This approach explicitly circumvents the statistical complexities of zero-inflated count regression, focusing instead on delivering actionable intelligence regarding whether regional authorities must be on high alert.

Accurately predicting this pre-ignition environment necessitates the sophisticated fusion of multimodal, high-resolution Earth observation data to capture the delayed effects of peatland drying over a 14-day antecedent window [4]. Crucial environmental inputs include continuous volumetric soil moisture and atmospheric metrics derived from ERA5-Land [6], delayed daily precipitation aggregates processed through CHIRPS v3 [7], and dynamic land cover probabilities extracted from Google Dynamic World [8]. Furthermore, Sentinel-1 Synthetic Aperture Radar (SAR) provides essential, cloud-penetrating measurements that serve as direct proxies for ground moisture and biomass [9]. By systematically comparing tabular baselines against advanced spatiotemporal architectures such as ConvLSTM and Temporal Transformers this study aims to isolate the predictive power of these combined environmental conditions from mere historical fire persistence.

2 Literature Review

2.1 Limitations of Optical and Thermal Satellite Detection

Conventional peatland fire detection relies on polar-orbiting satellites such as MODIS (Moderate Resolution Imaging Spectroradiometer) and VIIRS (Visible Infrared Imaging Radiometer Suite), which detect thermal anomalies on the Earth's surface. Although effective for large surface fires, this approach has three major limitations for peatland contexts.

First, MODIS has a 1 km spatial resolution per pixel, too coarse to detect small-scale *smoldering* fires common in the early stages of peat combustion [10]. Second, persistent cloud cover in tropical regions, especially during fire season, causes many detections to be missed by optical satellites. Third, peat fires burning below the surface have substantially lower surface temperatures than open flames, frequently falling below the detection threshold of MODIS algorithms [11]. Temporal hotspot patterns in Riau have been studied by [12], who identified correlations between hotspot sequences and peatland conditions, but that study did not integrate multimodal data. These limitations demonstrate that no single sensor modality can reliably predict peatland fires, motivating the need for a multimodal fusion approach.

2.2 Deep Learning Approaches for Fire Prediction

Advances in deep learning have opened new approaches for fire prediction beyond thermal anomaly detection. The Convolutional LSTM (ConvLSTM), introduced by [13] for precipitation nowcasting, has become a foundational architecture for various spatiotemporal prediction tasks, including wildfire prediction.

Several recent studies have demonstrated promising ConvLSTM performance for fire prediction. [14] developed an attention-based ConvLSTM to predict wildfire spread in the United States, showing that spatiotemporal deep learning significantly outperforms conventional Auto-Regressive Moving Average (ARIMA) models. [15] accelerated fire spread prediction using ConvLSTM for operational guidance, achieving accurate forecasts at a fraction of the computational cost of physics-based simulations.

2.3 Multimodal Fusion for Environmental Prediction

Multimodal fusion has become increasingly popular in environmental prediction because it captures complementary signals from multiple physical sources. For peatland fires, spatiotemporal fusion of radar, weather, rainfall, and land cover data remains largely unexplored in the literature. In particular, the integration of cloud-penetrating radar for direct soil moisture measurement alongside continuous atmospheric reanalysis and land cover information has not been evaluated for Indonesia's peatlands [16].

2.4 Research Gaps

Based on the literature surveyed above, several research gaps can be identified. First, no study has systematically compared tabular and spatiotemporal approaches for peatland hotspot prediction in Indonesia. Second, multimodal fusion integrating radar, weather, rainfall, land cover, and peat vulnerability maps has not been conducted in a spatiotemporal framework. Third, the relative contribution of environmental factors to peatland fire risk in Indonesia, analyzed through explainable AI methods, remains underexplored.

3 Methodology

3.1 Problem Formulation

The forecasting task is formulated as a binary spatiotemporal classification problem. Given observations from the previous 14 days ($t - 13$ until t), the objective is to predict whether a valid hotspot will occur during the following seven days ($t + 1$ until $t + 7$) for each 5-km grid cell.

A grid cell is labeled positive if at least two valid VIIRS hotspot detections occur within the prediction window. Binary classification is adopted because hotspot occurrence is highly imbalanced and zero-inflated, making early warning more operationally useful than hotspot count prediction.

3.2 Datasets Used

Six complementary datasets are integrated to represent different stages of wildfire development, as summarized in Table 1.

Table 1: Data sources used in this study.

Dataset	**Role**	**Temporal Type**
FIRMS VIIRS 375m	Hotspot labels	Daily
ERA5-Land	Meteorological variables	Daily
CHIRPS v3 (sat)	Rainfall	Daily
Sentinel-1 SAR	Soil moisture proxy	6–12 days
Dynamic World	Land-cover probabilities	2–5 days
Peatland Map	Static peat distribution	Static

CHIRPS-SAT is selected instead of CHIRPS-RNL to reduce redundancy with ERA5-Land, while Dynamic World is preferred over annual land-cover products because it provides probabilistic land-cover updates every 2–5 days.

3.3 Data Pre-processing

The administrative boundary of Riau Province was reprojected to the Indonesian national equal-area coordinate system (EPSG:9470) and overlaid with a 5 km $\times$ 5 km fishnet, yielding 84 columns $\times$ 90 rows (7,560 bounding-box cells). Cells whose center falls within the Riau polygon are designated as prediction targets; all cells, including those in neighboring provinces and water bodies that fall within the box, are retained as spatial context for the 15×15 convolutional patch.

Each dataset in Table 2 is independently resampled to the 5 km grid using its native aggregation logic (mean, median, count, nearest-neighbor, or rasterization). The resampled layers are then temporally stacked into a unified multi-channel table indexed by cell and date. VIIRS hotspot detections are aggregated per cell per day and converted to a binary label via the $k = 2$ persistence rule. Sliding windows of 14 input days and a 15×15 spatial patch are extracted to form the final tensor of shape $[N, 14, 15, 15, C]$, where $C \approx 22$ channels. The dataset is split temporally: all days in 2019–2022

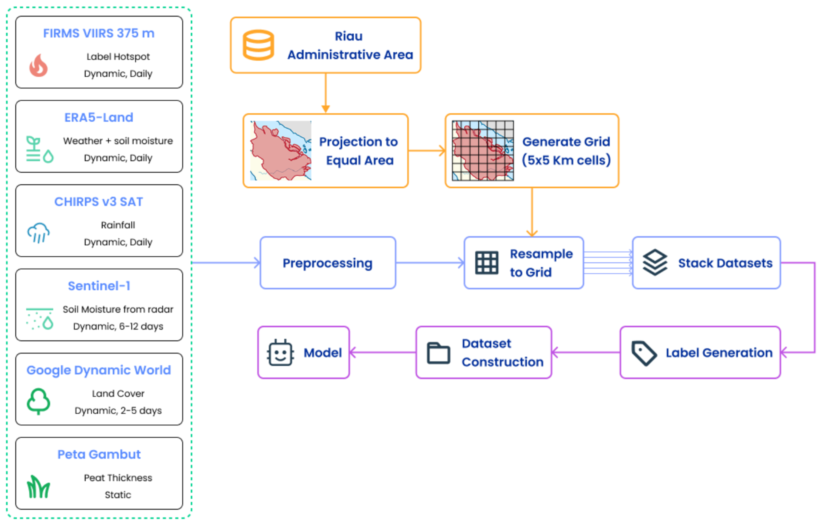

Figure 1: Data Pre-processing Pipeline

form the training set, and all days in 2023 form the held-out test set. Imputation and normalization are performed after splitting, with all statistics drawn exclusively from the training partition.

Table 2: Data Preprocessing Summary.

Dataset	Native format	Grid assignment	Preprocessing
FIRMS VIIRS 375 m	CSV point	Spatial join $\rightarrow$ daily count per cell	Confidence filter (nominal + high); $k = 2$ persistence over 7-day target $\rightarrow$ binary label
ERA5-Land	NetCDF, hourly, 9 km	Bilinear interpolation $\rightarrow$ 5 km	Aggregate hourly $\rightarrow$ daily: mean (t2m, d2m, u10, v10, swvl1, swvl2, ssr), sum (tp)
CHIRPS v3 SAT	GeoTIFF, daily, 5 km	Nearest-neighbor	—
Sentinel-1 SAR	GRD, 10 m, 6–12 day revisit	Median VV/VH per cell	Speckle filter; backscatter (dB); forward-fill gaps ≤ 8 days with binary availability mask
Dynamic World	GeoTIFF, 2–5 day, 10 m	Mean probability per class per cell	Exclude cloud-shadowed pixels; gap-fill missing days
Peat depth (BRGM FEG 2019)	Shapefile, static	Rasterize to 5 km	Parse numeric midpoint from string range (e.g., “1,5–2,0 m” $\rightarrow$ 1.75 m)

3.4 Model Development

Figure 2 summarizes the proposed prediction and decision-support workflow. Multimodal environmental observations are processed by the evaluated prediction models to generate hotspot forecasts. The prediction output is then combined with a Retrieval-Augmented Generation (RAG) module connected to a regulatory corpus, enabling the generation of context-aware recommendations aligned with Indonesian fire-management regulations.

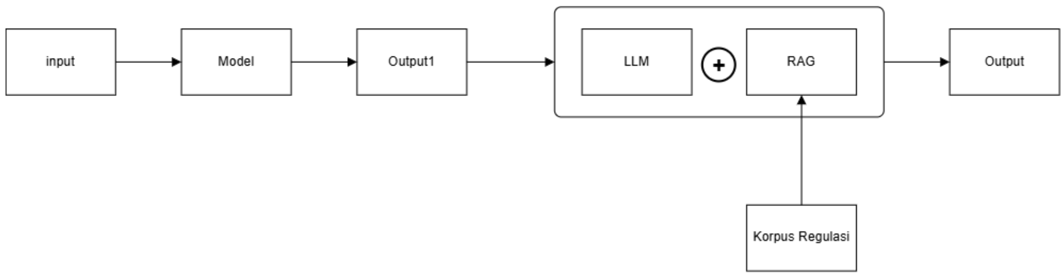

Figure 2: Prediction and decision-support workflow integrating hotspot forecasting with a regulatory-aware RAG module.

Four baseline models and two deep spatiotemporal models are evaluated, as summarized in Table 3.

Table 3: Evaluated models.

Category	**Model**	**Purpose**
Naive	Persistence	Reference baseline
Meteorological	Logistic Regression	Simple fire-danger baseline
Tabular	LR, RF, LightGBM	Machine learning baseline
Spatiotemporal	ConvLSTM	Main model
Spatiotemporal	Temporal Transformer	Main model

3.5 Model Evaluation

The primary evaluation follows a temporal split, where data from 2019–2022 are used for training and 2023 is reserved exclusively for testing.

Performance is primarily evaluated using PR-AUC, Recall, and F1-score, while ROC-AUC is reported as a secondary metric. Accuracy is not considered the primary evaluation metric because of the severe class imbalance.

To improve model interpretability, SHAP analysis is conducted for tabular models, whereas attention visualization is employed for the spatiotemporal architectures.

Several measures are implemented to prevent data leakage, including strict temporal separation between training and testing datasets, non-overlapping input and prediction windows, normalization using only training statistics, and exclusion of future hotspot information from all input features.

4 Conclusion

We present a pre-implementation research design for seven-day karhutla hotspot prediction in Riau, formulated as binary spatiotemporal classification on a 5 km grid. The design fuses six data sources across two feature regimes, applies a dual evaluation protocol along the temporal and spatial axes, enforces a ten-item leakage checklist, and states implementation considerations with explicit resolution criteria. The aim is a narrow but defensible scientific claim rather than a broad claim insufficiently supported by the experimental scope.

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